Lecture 2

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Causal Inference via Randomized Experiments

Potential Outcomes Framework

- In this lecture, we discuss the potential outcomes framework for analyzing causality and treatment effects.
- It offers a simple way to formalize causality as a mathematical concept.
- Consider the two *latent* (unobserved) variables

$$Y(1) \text{ and } Y(0).$$

  They represent the potential or counterfactual random outcomes for an observational unit when the unit is subject to treatment (treatment state $d = 1$) or no treatment (control or untreated state $d = 0$), see Rubin (1974).
- In an economic context, the treatment might be a training program or a policy intervention.

Counterfactuals

- It is also useful to introduce the potential response or structural function:

$$d \mapsto Y(d),$$

  which maps the potential treatment state $d \in \{0, 1\}$ to the random potential outcome $Y(d)$.
- The potential outcome $Y(d)(\omega)$ depends on the underlying state of the world ω. In our formalization, ω will represent randomness across observational units and from any other sources.
- The quantities $Y(1)$ and $Y(0)$ are "counterfactual" because they can't be simultaneously observed. That is, we generally do not have identical replicas of the observational units that are simultaneously subject to both treatment and control.
- This is also called the *fundamental problem* of causal inference.

Treatment Effects

- The **individual treatment effect** is

$$Y(1) - Y(0).$$

- This effect will vary across individuals as well as with other sources of randomness encoded in ω.
- It is generally unrealistic to uncover the individual treatment effect.
- Nevertheless, we can hope to estimate averages and the distribution of $Y(d)$ at the population level to compute quantities such as the **average treatment effect (ATE)**:

$$\delta = \mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)].$$

Consistency

- A commonly used assumption in causal inference is **consistency**:

$$Y := Y(D).$$

- The assumption make sure that the observed outcome is equal to $Y(1)$ after completion of a training program for a given person if she has completed the program (has $D = 1$), and is equal to $Y(0)$ if this person has not completed the training program (has $D = 0$).

Average Predictive Effect

Example 1

- Consider the following model:

$$
\begin{aligned}
Y(1) &:= \theta_1 + \epsilon_1 \\
Y(0) &:= \theta_0 + \epsilon_0 \\
D &:= 1(U > 0) \\
Y &:= Y(D),
\end{aligned}
$$

where θ_0 and θ_1 are constants, and $(\epsilon_0, \epsilon_1, U)$ are jointly normal random stochastic disturbances with mean 0 and covariance matrix Σ.

- In this example $\mathbb{E}[Y(1)] = \theta_1$, $\mathbb{E}[Y(0)] = \theta_0$, and the average treatment effect (ATE) is

$$\delta = \theta_1 - \theta_0.$$

- If **consistency** holds, we can identify the following **conditional averages**:

$$\mathbb{E}[Y \mid D = d] = \mathbb{E}[Y(d) \mid D = d], \text{ for } d \in \{0, 1\},$$

  directly from the population data.

- The difference of the two averages gives us the **average predictive effect** (APE) of treatment status on the outcome:

$$\pi = \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0].$$

  It measures the association of the treatment status with the outcome.

- While APE is identified – meaning computable from the population data – it may seem surprising (or not at all) that APE in general does not agree with the average predictive effect (ATE) δ:

$$\delta \neq \pi.$$

Selection Bias

- The difference between the APE and ATE is generally said to be due to *selection bias*.

Example 2

- Suppose we want to study the impact of smoking marijuana on life longevity.
- Assume that smoking marijuana has no causal effect on life longevity:

$$Y = Y(0) = Y(1), \text{so that } \delta = \mathbb{E}[Y(1) - Y(0)] = 0.$$

- However, the observed smoking behavior, D, is not assigned in an experimental study.
- The behavior is merely observed, and can be associated with poor health choices such as drinking alcohol, which are known to cause shorter life expectancy.
- In this case

$$\pi = \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0] < 0 = \delta.$$

- To sum up, in the smoking example, the observed "treatment" variable is potentially negatively associated with the potential health outcome, inducing the inequality $\mathbb{E}[Y(d)|D=1] < \mathbb{E}[Y(d)]$.
- It is useful to emphasize the main reason for having selection bias is that

$$\mathbb{E}[Y(d)|D=1] \neq \mathbb{E}[Y(d)]$$

  whenever D is not independent of $Y(d)$. If D and $Y(d)$ were independent,

$$\mathbb{E}[Y(d)|D=1] = \mathbb{E}[Y(d)]$$

  would hold since in this case D is uninformative about the potential outcome and drops out from the conditional expectation.
- The problem with observational studies like our contrived Example 2 is that the "treatment" variable D is determined by individual behaviors which may be linked to potential outcomes. This linkage generates selection bias - the disagreement between APE and ATE.

Randomized Control Trials

- The cleanest way to remove selection bias is through **random assignment** of treatment which leads to randomized control trials (RCT).

- **Assumption** (Random Assignment/Exogeneity): The treatment status is randomly assigned, namely D is statistically independent of each potential outcomes $Y(d)$ for $d \in \{0, 1\}$, which is denoted as

$$D \perp\!\!\!\perp Y(d)$$

  and $0 < \Pr(D = 1) < 1$.

- This assumption states that the treatment assignment mechanism is purely random, and ensures that there are units in treatment and in control.

- Does this hold in Example 1?

Theorem 1 (Randomization Removes Selection Bias]) Under random assignment, the average outcome in treatment group d recovers the average potential outcome under the treatment status d:

$$\mathbb{E}[Y \mid D = d] = \mathbb{E}[Y(d) \mid D = d] = \mathbb{E}[Y(d)],$$

for each $d \in \{0, 1\}$.

- Hence the average predictive effect and average treatment effects coincide:

$$\begin{aligned} \pi &:= \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0] \\ &= \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] =: \delta. \end{aligned}$$

- Because of this property, RCTs are considered golden standard in causal inference, and are routinely employed in a variety of important settings.
- Examples include evaluating the efficacy of medical treatment, vaccinations, training programs, marketing campaigns, and other kinds of interventions.

Statistical Inference

- Inference is based on the independent sample $(Y_i, D_i)_{i=1}^n$ obtained from the experiment, where index i denotes the observational unit.
- Estimation of two (conditional) means $\theta_d = \mathbb{E}[Y \mid D = d]$ for $d = 0$ and $d = 1$, can be done via linear regression or, equivalently, by considering two group means

$$\hat{\theta}_d = \frac{\mathbb{E}_n[Y1(D = d)]}{\mathbb{E}_n[1(D = d)]}.$$

- Under mild regularity conditions, we have that $\hat{\delta} = \hat{\theta}_1 - \hat{\theta}_0$ obeys

$$\sqrt{n}(\hat{\delta} - \delta) \xrightarrow{\mathrm{D}} N(0, V_{11} + V_{22}),$$

where

$$V = \begin{pmatrix} \frac{Var(Y1(D=0))}{P(D=0)} & 0 \\ 0 & \frac{Var(Y1(D=1))}{P(D=1)} \end{pmatrix}.$$

Example: Pfizer/BioNTech Covid Vaccine RCT

- Pfizer/BNTX was the first vaccine approved for emergency use in the EU and US to reduce the risk of Covid-19 disease.
- See the Food and Drug Administration (FDA) briefing for details about the RCT and the summary data.
- Volunteers were randomly assigned to receive either a treatment (2-dose vaccination) or a placebo, without knowing which they received, and the doctors making the diagnoses did not know whether a given volunteer received a vaccination or not.
- In other words, the trial was a double-blind randomized control trial.
- This notebook contains the analysis of the Pfizer-BioNTech Covid-19 Vaccine and Polio Vaccine RCTs.

Results

Efficacy Endpoint Subgroup	BNT162b2 N[a]=19965 Cases n1[b] Surveillance Time[c] (n2[d])	Placebo N[a]=20172 Cases n1[b] Surveillance Time[c] (n2[d])	Vaccine Efficacy % (95% CI)[e]
Overall	9 2.332 (18559)	169 2.345 (18708)	94.6 (89.6, 97.6)
Age group (years)			
16 to 17	0 0.003 (58)	1 0.003 (61)	100.0 (-3969.9, 100.0)
18 to 64	8 1.799 (14443)	149 1.811 (14566)	94.6 (89.1, 97.7)
65 to 74	1 0.424 (3239)	14 0.423 (3255)	92.9 (53.2, 99.8)
≥75	0 0.106 (805)	5 0.109 (812)	100.0 (-12.1, 100.0)

Figure 1: The aggregate data from the Pfizer RCT, source: FDA

- The trial had to be large, because the rate of Covid-19 infection was relatively low at the time.
- Specifically, the treatment group saw 9 Covid-19 cases per 19,965, while the control group saw 169 cases per 20,172.
- The estimated average treatment effect is about

$$-792.7$$

cases per 100,000, and the 95% confidence band is $[-922, -664]$.

Statistical Inference

- Sometimes we also have additional pre-treatment or pre-determined covariates W.
- We might be interested in either using these covariates to estimate average treatment effects more precisely or to describe heterogeneity of the treatment effects.
- For example, we might be interested in the impact of a treatment across age or income groups.

Regression and Statistical Inference for ATEs

- Here we make the strong linear assumption on the conditional expectation function

$$\mathbb{E}[Y \mid D, W] = D\alpha + X'\beta, \qquad (1)$$

  where $X = (1, W)$ contains an intercept and pre-treatment covariates W. This is clearly restrictive, but we will relax this below and makes things fully nonlinear later. The inferential result will be valid without this assumption.

- Using centered covariates implies that

$$\mathbb{E}[Y(0)] = \mathbb{E}[\mathbb{E}[Y \mid D = 0, X]] = \beta_1$$

  and

$$\mathbb{E}[Y(1)] = \mathbb{E}\mathbb{E}[[Y \mid D = 1, X]] = \beta_1 + \alpha.$$

- The average outcome in the untreated state is β_1, and the average treatment effect is α.

Estimator

- Equation 1 implies that

$$Y = D\alpha + X'\beta + \epsilon, \quad \epsilon \perp (D, X)$$

  implying that α coincides with the BLP coefficient in the BLP of Y on D and X.

- Application of the OLS theory in the previous chapter gives us under regularity conditions:

$$\sqrt{n}(\hat{\alpha} - \alpha) \xrightarrow{\mathrm{D}} N(0, V_{11}),$$

  with $V_{11} = \frac{\mathbb{E}[\epsilon^2 \tilde{D}^2]}{(\mathbb{E}[\tilde{D}^2])^2}$ where $\tilde{D} = D - \mathbb{E}[D]$ is the residual after partialling out linearly X from D.

The Role of Covariates

- One role of covariates in randomized experiments is "denoising" or improving the precision of estimation of quantities like the ATE. Without covariates, the OLS estimator $\bar{\alpha}$ estimates the projection coefficient α in the model:

$$Y = \alpha D + \beta_1 + U,$$

  with $\mathbb{E}[U(1, D)'] = 0$.
- The noise $U = X'\beta + \epsilon$ contains the "explainable" component $X'\beta$ if we were to include covariates.
- We then have that $\bar{\alpha}$ obeys

$$\sqrt{n}(\bar{\alpha} - \alpha) \xrightarrow{\text{D}} N(0, \bar{V}_{11}) \text{ with } \bar{V}_{11} = \frac{\mathbb{E}[U^2 \tilde{D}^2]}{(\mathbb{E}[\tilde{D}^2])^2}.$$

- Under our assumptions, it follows that $V_{11} < \bar{V}_{11}$ if $Var[X'\beta] > 0$.

Robust Standard Errors

- That is, pre-determined covariates improve the precision of estimating the ATE α.
- While the inferential result we derived is robust with respect to the linearity assumption on CEF (conditional expectation function), the improvement in precision itself is **not** guaranteed in general and hinges on the validity of the linearity assumption for the CEF.
- The following notebook constructs an example where controlling for predetermined covariates linearly lowers the precision (increases robust standard errors).

Treatment Effect Heterogeneity

- Next, we use covariates to describe heterogeneity of the treatment effects. For this purpose, we consider conditional average treatment effects (CATE):

$$\delta(W) = \mathbb{E}[Y(1) \mid W] - \mathbb{E}[Y(0) \mid W],$$

  which compare the average potential outcomes conditional on a set of covariates W.

- We can use RCTs to identify conditional average treatment effects by estimating the conditional predictive effects (CPE):

$$\pi(W) = \mathbb{E}[Y \mid D = 1, W] - \mathbb{E}[Y \mid D = 0, W].$$

- **Random Assignment Independent of Covariates**: Suppose that treatment status is randomly assigned, namely D is statistically independent of both the potential outcomes and a set of pre-determined covariates:

$$D \perp\!\!\!\perp (Y(d), W).$$

- This assumption spells out that, if we plan to use covariates in the analysis, randomization has to be made with respect to these covariates as well.
- In practice, it is often tempting to use post-treatment covariates, but the use of such variables runs the danger of violating random assignment independent of covariates.
- A common scenario where this may occur is when researchers encounter missing data from imperfect data collection in following-up with control and treated units to collect demographic information. When we drop observations with missing data, we implicitly condition on a post-treatment variable (missingness) which can cause violations of this assumption.
- The desire to assess randomization with respect to covariates motivates the following diagnostic procedure...

Testing Covariate Balance

- The random assignment assumption induces covariate balance. Namely, the distribution of covariates should be the same under both treatment and control:

$$W|D = 1 \sim W|D = 0,$$

  and, equivalently, $D|W \sim D$.

- A useful implication is that $\mathbb{E}[W|D = 1] = \mathbb{E}[W|D = 0]$. That is, D does not predict any covariate feature; and moreover, D is not predictable by W:

$$\mathbb{E}[D \mid W] = \mathbb{E}[D].$$

- These latter conditions are easily testable using regression tools.

Theorem 2 (Randomization with Covariates) Under *Random Assignment Independent of Covariates*, the expected value of Y conditional on treatment status $D = d$ and covariates W coincides with the expected value of potential outcome $Y(d)$ conditional on covariates W:

$$\mathbb{E}[Y \mid D = d, W] = \mathbb{E}[Y(d) \mid D = d, W] = \mathbb{E}[Y(d)|W],$$

for each d. Hence the conditional predictive and average treatment effects agree:

$$\pi(W) = \delta(W).$$

Discover Heterogeneity through CATE

- We can consider estimation of CATE through the lens of linear regression by use of an *interactive* model:

$$\mathbb{E}(Y \mid D, W) = DX'\alpha + X'\beta$$

with

$$X = (1, W')', \quad \mathbb{E}[W] = 0,$$

- Such a linear rule is called *interactive* because it includes the interaction (meaning, product) of D and W as a regressor, in addition to D and W. Here, we recover CATE via

$$\begin{aligned}\delta(W) = \mathbb{E}[Y(1) \mid W] - \mathbb{E}[Y(0) \mid W] = \\ \mathbb{E}[Y \mid D = 1, W] - \mathbb{E}[Y \mid D = 0, W] = X'\alpha.\end{aligned}$$

- Using that $\mathbb{E}[W] = 0$, the ATE is then $\delta = \mathbb{E}[\delta(W)] = \mathbb{E}[X'\alpha] = \alpha_1$, where α_1 is the first component of α.

- The function $W'\alpha_2$, where α_2 is the vector all elements of α excluding α_1, therefore describes the deviation of CATE away from the ATE.
- We can verify that α is the coefficient of the linear projection equation:

$$Y = \alpha' DX + \beta' X + \epsilon, \;\; \epsilon \perp (X, DX).$$

- Therefore, we can treat

$$\bar{D} := DX$$

  as a vector of technical treatments and invoke the "partialling out" approach for inference on components of α.
- Unlike the previous approach for ATE estimation, the "interactive" approach always delivers improvements in precision of estimating δ, even if the linearity assumption does not hold; this was demonstrated by Lin et al. (2013).

Reemployment Bonus RCT

- The notebook explores the use of covariates to improve precision and learn about heterogeneity in a Reemployment Bonus RCT.
- Here we re-analyze the Pennsylvania re-employment bonus experiment Bilias (2000), which was conducted in the 1980s by the U.S. Department of Labor to test the incentive effects of alternative compensation schemes for unemployment insurance (UI).
- In these experiments, UI claimants were randomly assigned either to a control group or one of five treatment groups. We focus our discussion on treatment group 4. In the control group the current rules of the UI applied.
- Individuals in the treatment groups were offered a cash bonus if they found a job within some pre-specified period of time (qualification period), provided that the job was retained for a specified duration; see the Penn Data Codebook for further details on the data.

- We consider the classical 2-sample approach, no adjustment (CL), the classical linear regression adjustment (CRA) and the interactive regression adjustment (IRA)
- Estimates of the ATE (est) on (log) unemployment duration and corresponding estimated standard errors (se) are given in the following table:

	CL	**CRA**	**IRA**
est	-0.08546	-0.07968	-0.07550
se	0.03586	0.03559	0.03560

- We see that the treatment group experiences an average decrease of about 7.8% in the duration of unemployment spells.
- The two regression estimators deliver estimates that are slightly more precise (have lower standard errors) than the simple difference in two means estimator.
- The regression estimators offer slightly lower estimates of the ATE. These differences likely occur due to *imbalances* in the treatment allocation.
- Though not reported here, there is not any statistically detectable heterogeneity in effects from looking at the IRA results given in the R Notebook for this example.

The limitations of RCTs

- The traditional formulation of Rubin's causal model relies on SUTVA. This assumption requires that the potential outcome observation of one unit should be unaffected by the assignment of treatments to other units, see Cox (1958).
- We consider some cases where this assumption might not hold:

1. In a vaccine example, this assumption holds if treatment and control populations are "small" subpopulations of the entire general population. If we vaccinate a sufficiently large percentage of people, reaching herd immunity, the outcomes for the control group would be essentially the same as outcomes for the treated SUTVA therefore would not hold.
2. In economics we refer to such spillover effects as externalities, and in some contexts, we may call them general equilibrium effects. For example, there is a positive externality created by people who take the vaccine (and people that don't take vaccine "free ride", once the vaccination level is high enough).

3. We might want to study the earning effect of getting a college degree versus not having a college degree. If treatment will target a relatively small subpopulation of people, there likely won't be any large general equilibrium wage effects. On the other hand, if the treatment will target a large subpopulation, the equilibrium wage will likely adjust (the college wage premium might decrease, for example).

Ethical, Practical and Generalizability Concerns

- Many RCTs are infeasible because implementing them would be unethical. The general ethical principles and guidelines for research involving human subjects are set out in the 1978 Belmont report "The Belmont Report: Ethical Principles and Guidelines for the Protection of Human Subjects of Research" (1978).
- For example, we previously considered a hypothetical RCT where individuals are assigned to a smoking treatment group. The trial would violate the principle of "beneficence" as the researcher might be causing physical harm to study participants by assigning them to smoking.
- Thus, RCTs are rarely a feasible means of retrieving the causal effects of harmful interventions as they tend to be unethical.
- RCTs may also face practical issues. They can be prohibitively expensive where the treatment is costly, data collection costs are high or the sample size required for adequate power is high.

- Even in the best case, where an RCT is successfully implemented and we are confident in our retrieved average treatment effect, it may be difficult to generalize (or extrapolate) the result of an RCT in a specific context to a general finding. This difficulty might be because local conditions or implementation capacity materially differ between where interventions are staged or because the scale of the intervention is important.

Concluding Notes

- RCTs have had a profound influence on business, economics and science more generally. For example, RCTs are routinely used to study the efficacy of drugs and efficacy of various programs in labor and development economics, among other subfields of economics.
- The FDA moved to RCTs as the gold standard of proving that treatments work in 1970s-80s.
- In the tech industry and marketing, RCTs are also called "A/B Testing" and are now widely used. Many major tech companies have their own experimental platforms to carry out thousands of experiments. See, for example, the ExP platform at Microsoft and the WebLab platform at Amazon.
- The expansion of the use of experimentation in economics is associated with the work of Abhijit Banerjee, Esther Duflo, and Michael Kremer, the recipients of Alfred Nobel Memorial Prize in Economics for their experimental approach to alleviating global poverty, see here.

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